Meet Udeshi

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Ph.D. Candidate, NYU Tandon School of Engineering — Expected graduation: May 2027

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SUMMARY

Cyber-physical systems security researcher in a control and robotics lab, who builds digital twins for malware analysis and reverse-engineers controller binaries. Experience with LLM agentic automation for offensive security. Background in ML compiler engineering at Qualcomm, and hardware security at IIT Bombay.

EDUCATION

NYU Tandon School of Engineering
Ph.D., Electrical Engineering — Sep 2022–present
Control/Robotics Research Lab, Center for Cybersecurity. GPA: 3.958/4
Advisors: Prof. Farshad Khorrami and Prof. Ramesh Karri

Indian Institute of Technology Bombay (IIT Bombay)
B.Tech. and M.Tech. (Dual Degree), Electrical Engineering — July 2014–June 2019
Computer Architecture and Dependable Systems Lab. CGPA: 8.18/10
Thesis Advisor: Prof. Virendra Singh

RESEARCH PROJECTS

Cyber-Physical Systems Security — Sep 2025–present
Ph.D. research, Control/Robotics Research Lab, NYU

- Built a sandbox-enabled digital twin of a power grid, so attacks can be tested in a realistic, system-aware setting without risk to physical hardware (IOLTS 2026).
- Built SaMOSA, a side-channel observation platform for CPS malware on OpenPLC runtimes over Modbus TCP: an orchestrated QEMU sandbox with ARM64, x86-64 and PPC64 guests that records performance counters, disk, network and syscall activity (IEEE ESL / EMSOFT 2026).
- Designed an LLM binary-diffing framework for software supply chain security that detects malicious code in software updates, with LLM inference and batch completion via vLLM on parallel HPC jobs.
- Prototyped *Ransomware 3.0/PromptLock* to evaluate LLM-powered ransomware lifecycles with Lua code generation and stage-wise attack chain execution. Covered by NYU Tandon News (2025) and MIT Technology Review (2026).

Reverse Engineering of Cyber-Physical Systems — Aug 2022–present
Ph.D. research, Control/Robotics Research Lab, NYU

- Recovered control laws from compiled controller code with REMaQE, an `angr` symbolic execution framework with 100% accuracy on C and Simulink-generated binaries (ACM TCPS).
- Designed REMEND, a Transformer neural decompiler for equations (ACM TIST). One model covers three ISAs, three optimization levels and two languages, with 89.8–92.4% accuracy on C and Fortran, 8% above state-of-the-art on real-world code, and 0.13 s per function.
- Trained the REMEND Fairseq Transformers (up to 12M parameters) on the NYU HPC.
- REMaQE found a bug in the Linux `tmon` PID controller; submitted a kernel patch (2023).

Tamper-Proof Network Measurements on a SmartNIC
Ph.D. research, Control/Robotics Research Lab (IEEE TNSM 2025)
Designed tamper-proof traffic measurement for intrusion detection on a Netronome SmartNIC (P4, embedded C), out of reach of host-OS rootkits. Two designs (associative table, count-min sketch) run at 10 Gbps with 99.5% of baseline throughput and 12–23 µs added latency.

Hardware Security (Masters Thesis) — Aug 2018–June 2019
CADSL, IIT Bombay. Advisor: Prof. Virendra Singh
Designed cache side-channel attacks that disable the hardware prefetcher to strip its noise (against ciphers) and that leak between SMT threads through the shared re-order buffer (via IPC). Built an LLVM-IR framework for x86-to-ARM program migration in the heterogeneous-ISA core HIDC.

LLM Agents for Cybersecurity Mar 2024–present
Ph.D. research, Center for Cybersecurity, NYU

- Curated NYU CTF Bench (200 challenges, NeurIPS 2024) with Docker-hosted challenges and evaluations, evaluating five state-of-the-art LLMs, and co-authored a lightweight CTF benchmark with LLM-as-a-judge scoring (AAAI 2026). Developed EnIGMA (ICML 2025), whose Interactive Agent Tools give LLM agents debuggers and other interactive utilities, with state-of-the-art results on NYU CTF Bench and InterCode-CTF over 350+ challenges.
- Developed D-CIPHER, a planner–executor multi-agent framework with state-of-the-art solve rates of 22% on NYU CTF Bench and 20% on Cybench. Organized the CSAW LLM Attack Challenge (2024–2026), judging agentic solutions and agent frameworks from 64 teams over three years.

WORK EXPERIENCE

Siemens Healthineers USA, Princeton, NJ May 2026–Aug 2026
AI Cyber Intern, AI Factory

- Modeled the cyber-clinical impact of unpatched vulnerabilities in clinical and hospital critical infrastructure, toward ARPA-H UPGRADE's goal of more efficient vulnerability patching.
- Contributed to LLM-based generation of synthetic vulnerabilities for hospital devices and networks, as training data for an RL vulnerability-patching agent.
- Ran multi-GPU vLLM inference on single nodes, and multi-node, multi-CPU hospital simulations as dependency-linked Slurm jobs.

Qualcomm R&D, Bangalore, India July 2019–July 2022
Senior Engineer, ML Compiler Team for Cloud AI100 Accelerator

- Optimized autonomous-driving, 2D/3D computer vision and recommendation models for the Cloud AI100 accelerator through graph optimizations, and deployed a power-efficient KCF object-tracking pipeline on AI100.
- Implemented accelerator kernels for the input and output processing stages inside ResNet and YOLO graphs; built multi-core, multi-thread data tiling, memory management, graph scheduling and operator fusion in the compiler.
- Worked deep in PyTorch and TensorFlow graph-capture and op-lowering internals, contributed to the open-source *PyTorch Glow* compiler, and explored polyhedral loop and array optimization alongside Halide-style scheduling.

Amazon India Jul 2017–Aug 2017
Software Development Intern, Transportation Financial Systems
Processed and sorted 1 million+ receipts daily (DynamoDB, SQS); automated setup of 30+ AWS resources with CloudFormation.

Google Summer of Code, Kivy May 2016–Aug 2016
Python Native UI Framework Game Engine
Built a Python/Cython Tiled maps module and a Cython-optimized entity-component animation system for the KivEnt game engine.

SELECTED PUBLICATIONS

Equal contributions are marked with †.

[C1] **M. Udeshi**†, M. Raz†, P. Krishnamurthy, R. Karri, and F. Khorrami. "Sandbox-Enabled Digital Twin for Cyber-Physical Systems." *IEEE International Symposium on On-Line Testing and Robust System Design (IOLTS)*, 2026.

[C2] **M. Udeshi**, V. S. C. Putrevu, P. Krishnamurthy, R. Karri, and F. Khorrami. "An Orchestrated Sandbox for Side-Channel Analysis of Cyber-Physical Systems Malware." *IEEE Embedded Systems Letters*, presented at *Embedded Software Conference (EMSOFT)*, 2026.

[C3] T. Abramovich, **M. Udeshi**†, M. Shao†, K. Lieret, H. Xi, K. Milner, S. Jancheska, J. Yang, C. E. Jimenez, F. Khorrami, P. Krishnamurthy, B. Dolan-Gavitt, M. Shafique, K. Narasimhan, R. Karri, and O. Press. "EnIGMA: Interactive Tools Substantially Assist LM Agents in Finding Security Vulnerabilities." *International Conference on Machine Learning (ICML)*, 2025.

[C4] M. Shao†, S. Jancheska†, **M. Udeshi**†, B. Dolan-Gavitt†, H. Xi, K. Milner, B. Chen, M. Yin, S. Garg, P. Krishnamurthy, F. Khorrami, R. Karri, and M. Shafique. "NYU CTF Bench: A Scalable Open-Source Benchmark Dataset for Evaluating LLMs in Offensive Security." *Conference on Neural Information Processing Systems (NeurIPS)*, 2024.

[C5] M. Shao, N. Rani, K. Milner, H. Xi, **M. Udeshi**, S. Aggarwal, V. S. C. Putrevu, S. K. Shukla, P. Krishnamurthy, F. Khorrami, R. Karri, and M. Shafique. "Towards Effective Offensive Security LLM Agents: Hyperparameter Tuning, LLM as a Judge, and a Lightweight CTF Benchmark." *AAAI Conference on Artificial Intelligence*, 40(35):29660–29668, 2026.

[J1] **M. Udeshi**, P. Krishnamurthy, H. Pearce, R. Karri, and F. Khorrami. "REMaQE: Reverse Engineering Math Equations from Executables." *ACM Transactions on Cyber-Physical Systems*, 8(4), 2024.

[J2] **M. Udeshi**, P. Krishnamurthy, R. Karri, and F. Khorrami. "REMEND: Neural Decompilation for Reverse Engineering Math Equations from Binary Executables." *ACM Transactions on Intelligent Systems and Technology*, 17(3), 2025.

[J3] **M. Udeshi**, P. Krishnamurthy, R. Karri, and F. Khorrami. "Tamper-Proof Network Traffic Measurements on a NIC for Intrusion Detection." *IEEE Transactions on Network and Service Management*, 22(2), 2025.

HONORS AND AWARDS

DAC Young Fellowship — Design Automation Conference, 2023

Student Travel Grant to present at ESWEEK (EMSOFT) — IEEE CEDA, 2026

RoCStar (Recognition of Outstanding Contributions), AI100 compiler and KCF — Qualcomm, 2021

Gold Medal, Indian National Physics Olympiad (top 35 nationwide) — HBCSE, 2014

Tech Special Mention, among 500+ hostel residents — IIT Bombay, 2016

SERVICE AND PEER REVIEW

Peer Review: AAAI 2026, AAAI 2027, ICLR 2026, ICML 2026, NeurIPS 2026, CPS-IoTSec 2025, *ACM Transactions on Intelligent Systems and Technology*, *The Computer Journal*, *The Journal of Supercomputing*, *Journal of Computer Virology and Hacking Techniques*.

2016–2017	Manager, Electronics Club	IIT Bombay
2015	Reviewer, 46th International Physics Olympiad	Mumbai, India
2014	Volunteer tutor, Supplemental Learning Program	Vidya NGO Mumbai

TEACHING AND MENTORING

2020–2021	Mentor, hardware security project, Qualcomm Innovation Fellowship.	Qualcomm
Spring 2019	Teaching Assistant, VLSI Design Lab (Prof. Sachin Patkar).	IIT Bombay
Fall 2018	Teaching Assistant, Microprocessors (Prof. Virendra Singh).	IIT Bombay
Summer 2016	Teaching Assistant, CS101 Computer Programming (Prof. D. B. Phatak).	IIT Bombay

OPEN SOURCE

Linux kernel (`tools/thermal` patch), Kivy (KivEnt, GSoC 2016), beets, fast_playlist.

SKILLS

CPS, embedded	Embedded C/C++, MATLAB/Simulink, OpenPLC, Modbus TCP, Linux, threat modeling
Security	Symbolic execution (`angr`), static/dynamic binary analysis (Binary Ninja, Ghidra, `radare2`), neural decompilation (Fairseq), binary diffing (`ghidriff`), malware sandboxing (QEMU), side-channel analysis (gem5, PIN)
Programming	Python, C, C++, Shell/Bash, Assembly, Verilog, JavaScript, HTML, Java
ML and infra	PyTorch, TensorFlow, Fairseq, vLLM, ReAct LLM agents, PyTorch Glow, LLVM, polyhedral optimization, Docker, Slurm/HPC
Coursework	Hardware Security & Trust, Deep Learning, Advanced Computer Architecture, Advanced Hardware Design, Computer Vision, VLSI Design